Data Types in Statistics: A Beginner's Guide

Learning objectives: Learner should be able to:

- Distinguish between qualitative (categorical) and quantitative (numerical) variables.
- Classify qualitative variables as nominal or ordinal.
- Classify quantitative variables as discrete or continuous.
- Identify the measurement scale of a variable as nominal, ordinal, interval, or ratio.
- Select appropriate summary statistics and graphical displays for different types of data.
- Apply these classifications to a small dataset and justify each decision.

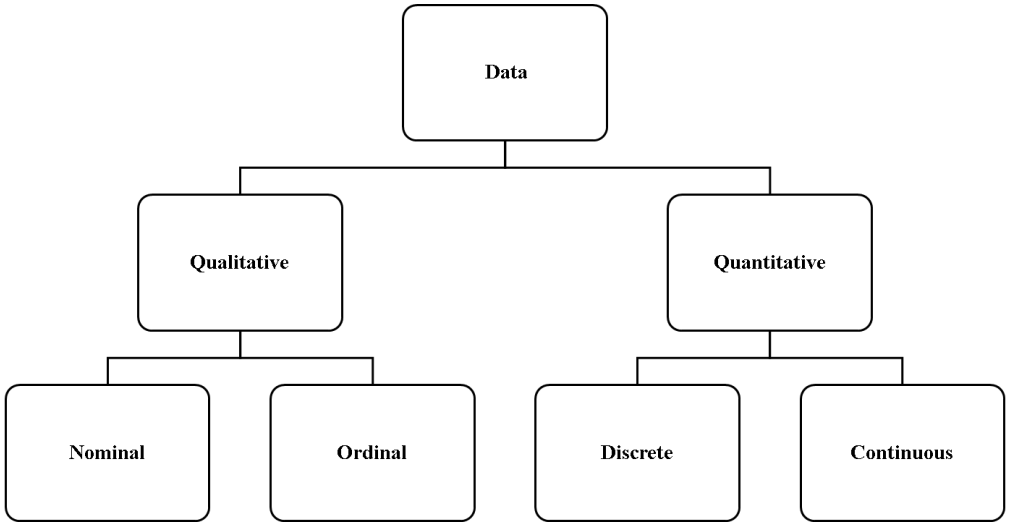
Introduction

Data are the raw information collected in research, business, healthcare, education and everyday life. Before analysing data, it is important to identify the type of data because this determines the appropriate graphs, summary statistics, and statistical tests.

Data are commonly classified as qualitative or quantitative. They can also be described using four measurement scales: nominal, ordinal, interval, and ratio. Understanding these classifications helps researchers choose valid methods and interpret results accurately.

For example, a graduate student studying the relationship between study hours and examination scores must distinguish between categorical variables, such as gender or programme, and numerical variables, such as age, marks, or study hours.

Classification by Nature of Data



Classification	Type	Description	Examples

Categorical data/ Qualitative	Nominal	Categories with no natural order or ranking	Gender, blood group, nationality, department
Categorical data/Qualitative	Ordinal	Categories with a meaningful order, but unequal gaps between categories	Education level, satisfaction rating, academic year
Numerical data/ Quantitative	Discrete	Countable values with no intermediate values	Number of books, number of survey responses, number of children
Numerical data/ Quantitative	Continuous	Measured values that can take any value within a range	Height, weight, time, temperature

Qualitative Data

Qualitative data, also called categorical data, describe qualities, characteristics, or categories. The values are usually non-numerical, although categories may be assigned numerical codes for computer analysis.

There are two main types of qualitative data:

- Nominal data: Categories with no natural order or ranking.
- Ordinal data: Categories with a meaningful order, but the distance between categories is not necessarily equal.

Nominal Data

Nominal data consist of categories that cannot be logically arranged from smallest to largest or lowest to highest.

- Categories are mutually exclusive.
- There is no inherent order.
- The mode and frequency counts are appropriate summaries.

Examples include gender, blood group, nationality, marital status, hair colour, and university department.

Graduate-student examples: field of study, such as Economics, Psychology, Engineering, or Medicine; preferred research method, such as survey, interview, or experiment; and type of accommodation, such as hostel, rented flat, or family home.

Ordinal Data

Ordinal data are categorical data with a meaningful order or ranking. However, the differences between adjacent categories cannot be assumed to be equal.

- Categories have a clear sequence.
- Differences between categories are not necessarily uniform.
- The median and mode can be used; non-parametric tests are often appropriate.

Examples: education level, satisfaction ratings, socioeconomic status, and Likert-scale responses.

Graduate-student examples: academic year, such as first year, second year, or final year; satisfaction with library facilities rated as poor, fair, good, or excellent; and research-skill level rated as beginner, intermediate, or advanced.

Quantitative Data

Quantitative data are numerical data that can be measured or counted. They are suitable for mathematical and statistical analysis, including calculation of means, medians, standard deviations, correlations, and regression models.

Quantitative data are divided into discrete and continuous data.

Discrete Data

Discrete data can take only specific, distinct values, usually whole numbers. These values are typically obtained by counting.

- Values are countable.
- There are no intermediate values between two adjacent observations.
- Bar charts, dot plots, and frequency distributions are commonly used.

Examples include the number of students in a class, number of cars in a car park, number of children in a family, and number of errors in an assignment.

Graduate-student examples: number of research papers published, number of courses completed, number of survey responses received, and number of absences in a semester.

Continuous Data

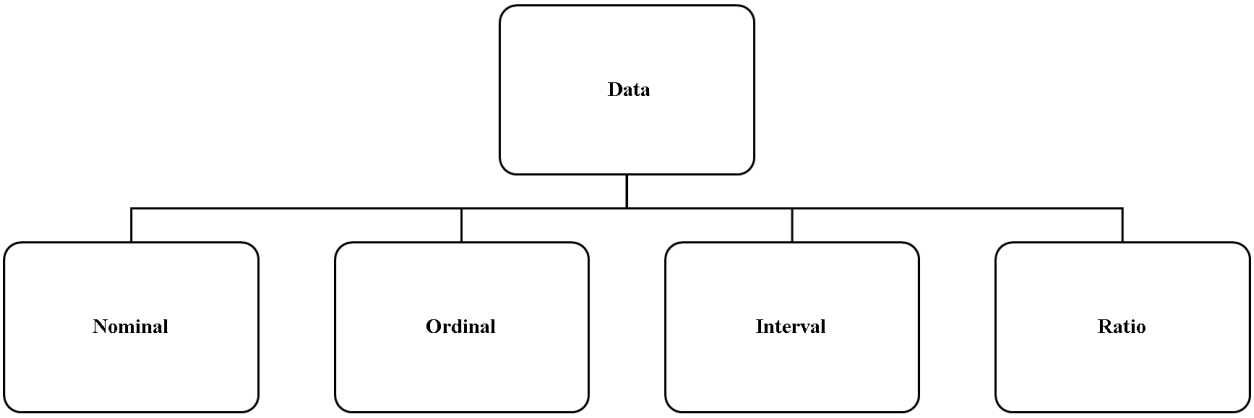
Continuous data can take any value within a given range, including decimal values. These values are usually obtained by measuring.

- Values are measured rather than counted.
- There can be infinitely many possible values within a range.
- Histograms, line graphs, box plots, and regression analysis are commonly used.

Examples include height, weight, temperature, time, distance, and blood pressure.

Graduate-student examples: time spent studying per day, body mass index, reaction time in an experiment, monthly income, and examination score expressed as a percentage.

Classification by Measurement Scale



Measurement Scales

Measurement scales describe the level of information contained in a variable. The four scales are nominal, ordinal, interval, and ratio.

Type	Description	Examples
Nominal	No order, no equal intervals, no true zero	Blood group, nationality
Ordinal	Ordered categories, but no equal intervals or true zero	Satisfaction level, education level
Interval	Equal intervals, but no true zero	Temperature in °C or °F, IQ score
Ratio	Equal intervals and a true zero	Height, weight, age, income

Nominal Scale

A nominal scale classifies observations into unordered categories. Examples include gender, blood group, nationality, and employment status.

Suitable summaries include frequencies, percentages, and the mode.

Ordinal Scale

An ordinal scale classifies observations into ordered categories. Examples include education level, customer-satisfaction rating, and disease severity.

Suitable summaries include the median, mode, percentiles, and certain non-parametric tests.

Interval Scale

An interval scale has equal intervals between values but no true zero point. A value of zero does not mean complete absence of the characteristic being measured.

Examples include temperature in Celsius or Fahrenheit and IQ scores. Differences between values are meaningful, but ratios are not.

For example, 30 degrees Celsius is not twice as hot as 15 degrees Celsius because zero degrees Celsius does not represent the absence of temperature.

Ratio Scale

A ratio scale has equal intervals and a true zero point. A value of zero indicates the complete absence of the characteristic being measured.

Examples include height, weight, age, income, distance, and time. All arithmetic operations are meaningful, including ratios.

For example, a person who is 180 cm tall is twice as tall as a person who is 90 cm tall.

Comparison Table

Scale	Order	Equal Intervals	True Zero	Examples
Nominal	No	No	No	Gender, blood group, nationality
Ordinal	Yes	No	No	Education level, satisfaction rating
Interval	Yes	Yes	No	Temperature in Celsius or Fahrenheit, IQ score
Ratio	Yes	Yes	Yes	Height, weight, age, income

Worked Examples

Example 1 : Classifying variables in Student Dataset.

A researcher collects the following information from 100 graduate students. The following table displays the first five observations from an illustrative dataset of 100 students, Identify the data type, measurement scale and appropriate summaries for each variable.

Student	Major Elective Subject	Satisfaction with library	Number of books read	Study hours per week
1	Accounts	Good	5	10
2	Finance	Excellent	8	16
3	Statistics	Fair	3	9.5
4	Management	Poor	2	7.5
5	Accounts	Good	7	14.5
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Answer:

Variable	**Data type**	**Measurement scale**	**Appropriate summaries**
Major Elective Subject	Qualitative	Nominal	Frequencies, percentages, mode
Satisfaction with library	Qualitative	Ordinal	Frequencies, median, mode
Number of books read	Quantitative, discrete	Ratio	Mean, median, standard deviation
Study hours per week	Quantitative, continuous	Ratio	Mean, median, standard deviation, histogram

Example 2: Choosing an Appropriate Summary

A survey records the preferred learning method of 200 postgraduate students: online lectures, classroom lectures, or blended learning.

Because the variable is nominal, the researcher should report frequencies and percentages. For example, 90 students preferred online lectures, 70 preferred classroom lectures, and 40 preferred blended learning. The mode is online lectures.

It would not be appropriate to calculate a mean learning method because the categories cannot be meaningfully averaged.

Example 3: Choosing an Appropriate Graph

A graduate student measures the daily study time of 100 students, in hours. The data range from 0.5 to 9.8 hours.

Because study time is continuous quantitative data, a histogram or box plot is appropriate. A histogram can show whether study time is symmetric, skewed, or contains outliers, whereas box plot is more useful for systematically flagging observations using an outlier rule.

A pie chart would not be suitable unless the study time is first grouped into categories, such as less than 2 hours, 2 to 4 hours, and more than 4 hours.

Practical Questions

The following questions can be used to classify variables in a research project:

1. Is the variable numerical or categorical?
2. If it is categorical, do the categories have a natural order?
3. If it is numerical, is the variable counted or measured?
4. Does the variable have a true zero point?
5. Which graph or summary statistic is most appropriate for the variable?
6. Which statistical test is suitable for the research question?

Practice Questions

Identify the Data type and the measurement scale in the following examples:

1. Blood group of a patient.
2. Satisfaction rating of a university canteen: poor, fair, good, or excellent.
3. Number of research articles read by a student in one week.
4. Weight of a laboratory sample, in grams.
5. University rank in a national ranking system.
6. Temperature recorded in a chemistry laboratory, in degrees Celsius.
7. Monthly stipend received by a research scholar, in rupees.
8. Level of agreement with a survey statement: strongly disagree, disagree, neutral, agree, or strongly agree.
9. A researcher collects the following information from 50 graduate students:

    Department: Economics, Psychology, Engineering, Medicine.
    Satisfaction with campus Wi-Fi: Poor, Fair, Good, Excellent.
    Number of books borrowed from the library in one month.
    Time taken to complete an online examination, in minutes.

    Identify the data type, measurement scale and appropriate summaries for each variable.

Hint [Department is nominal qualitative data because the categories have no order.

Satisfaction rating is ordinal qualitative data because the categories are ordered, but the gaps between them are not equal.
Number of books borrowed is discrete quantitative data because it is counted in whole numbers.
Time taken is continuous quantitative data because it is measured and can include decimal values]

Answer Hints:

Q.no.	Variable	Data type	Scale
(1)	Blood group	Qualitative	Nominal
(2)	Satisfaction rating	Qualitative	Ordinal
(3)	Number of articles read	Quantitative, discrete	Ratio
(4)	Weight in grams	Quantitative, continuous	Ratio
(5)	Rank	Qualitative	Ordinal
(6)	Temperature in °C	Quantitative, continuous	Interval
(7)	Monthly stipend	Quantitative, continuous	Ratio
(8)	Survey statement	Qualitative	Ordinal
(9)	Department is nominal qualitative data because the categories have no order. Satisfaction rating is ordinal qualitative data because the categories are ordered, but the gaps between them are not equal. Number of books borrowed is discrete quantitative data because it is counted in whole numbers. Time taken is continuous quantitative data because it is measured and can include decimal values		

Note: income and examination scores are sometimes treated as continuous for statistical modelling, but recorded values may be discrete because of measurement or reporting precision.

Key Takeaways

- Qualitative data describe categories, while quantitative data represent numerical measurements or counts.
- Nominal data have no order, whereas ordinal data have a meaningful order.
- Interval data have equal intervals but no true zero, whereas ratio data have equal intervals and a true zero.
- Discrete data are counted, while continuous data are measured.
- The type of data determines the appropriate graph, summary statistic, and statistical test.
- Correct classification of data is essential for valid analysis and meaningful research conclusions.